2024-25 First Semester
MATH1053 Linear Algebra I

Assignment 8

Due Date: Due on 26/Nov/2024 (Tuesday), at the end of lecture.

- Write down your **CHN name** and **student ID**. Write neatly on **A4-sized** paper (*staple if necessary*).
- **Late submissions or answers without steps will not be graded.**

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1. Determine if the following vectors are linearly independent, and justify your answers:
   (a) $\begin{pmatrix}1\\0\\0\end{pmatrix}, \begin{pmatrix}0\\1\\1\end{pmatrix}$ in $\mathbb{R}^3$
   (b) $\begin{pmatrix}2\\1\\-2\end{pmatrix}, \begin{pmatrix}3\\2\\-2\end{pmatrix}, \begin{pmatrix}2\\2\\0\end{pmatrix}$ in $\mathbb{R}^3$
   (c) 2, x^2, x, $2x+3$ in P_3
2. Let $A \in \mathbb{R}^{m\times n}$. Show that if A has linearly independent column vectors, then $N(A) = \{\mathbf{0}\}$.
3. Let $\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_k$ be linearly independent vectors in $\mathbb{R}^n$, and let A be a nonsingular $n \times n$ matrix. Show that $A\mathbf{x}_1, \cdots, A\mathbf{x}_k$ are linearly independent.
4. Show that any finite set of vectors that contains the zero vector must be linearly dependent.
5. Let $\mathbf{v}_1$, and $\mathbf{v}_2$ be two vectors in a vector space V. Show that $\mathbf{v}_1$ and $\mathbf{v}_2$ are linearly dependent if and only if one vector is a scalar multiple of the other.
6. Let W be the set of all symmetric matrices in $\mathbb{R}^{2\times 2}$.
   (a) Show that W is a subspace of $\mathbb{R}^{2\times 2}$.
   (b) Compute $\dim W$.
7. Write down a basis for the null space of matrix in HW7-Q7, respectively. (No further steps are needed.)
8. Determine the dimension of the following subspaces W by finding a basis
   (a) W is the set of all vectors of the form $(a+b, a-b+2c, b, c)^T$, where $a, b, c \in \mathbb{R}$.
   (b) $W = \{(s+4t, t, s, 2s-t) : s \text{ and } t \text{ are real numbers.}\}$

No need to hand in the MATLAB exercises. **Tab F1 to get help** from MATLAB if needed.

9. (MATLAB exercise) Building matrices and compute the determinants

   (a) Define a square matrix by yourself and compute the determinant by hand;

   (b) use the command "det( )" in MATLAB to verify your answer in part(a). For example,
   >> A = [1 2 3 2 1;0 0 0 1 -1;1 2 0 -1 1;3 2 1 2 3; 1 0 0 0 0]
   >> det(A)

   (c) Get familiar with the commands "eye(k)", "ones(m,n)", "zeros(m,n)", "magic(k)", "diag", "rand" and "rref".

10. (MATLAB exercise) Building matrices using partitioning

    (a) Matrices in MATLAB can be built up by patching together smaller matrices into a big matrix. The smaller matrices must fit together exactly along rows and columns and not leave any spaces unfilled. For example type the following in MATLAB:
    A=[1,2,3;3,2,1]
    B=[7;8]
    C=[4,5,6]
    D=[A,B;C,0]
    E=[A',zeros(3,7);eye(2,6),A]
    All these cases should have worked fine. Try to understand why.

    (b) Create the following matrix A in MATLAB by defining A1=[1,2;3,4] and A2=3*eye(3) first, then using "blkdiag".

$$A = \begin{bmatrix} 1 & 2 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 3 \end{bmatrix}$$

11. (MATLAB exercise) Upper(lower) triangular matrix and LU factorization

    (a) Get familiar with the commands "triu", "tril", "lu".

    (b) Create a matrix A=magic(4) and see what are triu(A) and tril(A)?

    (c) Can A=ones(4,4) be LU factorized? If yes, find L and U.